Project Design Phase-II
Technology Stack (Architecture & Stack)

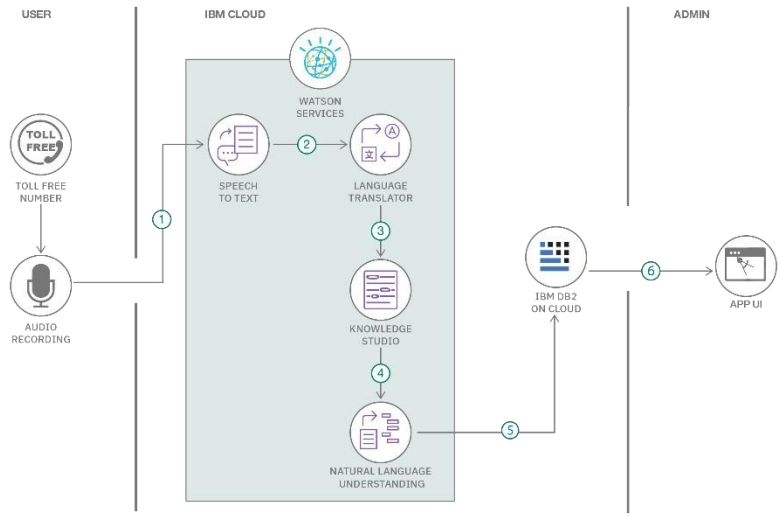
Date	11 October 2026
Team ID	ServiceNow - A2
Project Name	Car Rental Request Automation in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Employee-facing catalog item and request form.	ServiceNow Service Catalog / Catalog Item
2	Request Management	Creates and tracks the service request and Requested Item (RITM).	ServiceNow Request Management
3	Automation Logic	Controls validation, approval, assignment and fulfillment stages.	Flow Designer / Workflow
4	Approval Management	Routes requests to the appropriate approver and records the decision.	ServiceNow Approvals
5	Assignment & Fulfillment	Routes approved requests to the Transport team and records fulfillment information.	Assignment Groups / RITM
6	Notifications	Sends request and status updates to relevant users.	ServiceNow Notifications
7	Data Storage	Stores request, variable, approval, assignment and task information.	ServiceNow Platform Database
8	Access Control	Controls access using roles, ACLs and platform permissions.	ServiceNow Roles / ACLs
9	Deployment	Moves configuration between instances for testing and deployment.	Update Sets

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required for the core solution; ServiceNow platform capabilities are used.	ServiceNow
2	Security Implementations	Role-based access, ACLs, authentication and controlled access to catalog, request and fulfillment records.	ServiceNow Roles / ACLs
3	Scalable Architecture	Supports additional request variables, approval stages and fulfillment rules as demand grows.	ServiceNow Platform
4	Availability	Cloud-hosted ServiceNow platform provides access to the catalog and request records according to platform availability.	ServiceNow
5	Performance	Use mandatory fields, efficient flow conditions and targeted notifications to minimize unnecessary processing.	Flow Designer / ServiceNow

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d

Technical Architecture – Car Rental Request Automation

The architecture uses ServiceNow as the central platform. The employee interacts with the Service Catalog; ServiceNow creates and manages the request/RITM, executes Flow Designer or Workflow automation, routes approvals and assignments, stores request information and sends notifications.

High-Level Architecture Flow

EMPLOYEE / REQUESTER ↓ ServiceNow Service Catalog – Car Rental Request
↓ Catalog Variables → Request / Requested Item (RITM)
↓ Flow Designer / Workflow ➔ Validation → Approval → Assignment
↓ Transport Team * Availability Check → Fulfillment Details → Completion
↓ Notifications & Status Tracking ➔ Requester / Approver / Transport Team

Table-1: Components & Technologies

Table-2: Application Characteristics

References

- ServiceNow Service Catalog documentation
- ServiceNow Flow Designer documentation
- ServiceNow Request / Requested Item (RITM) documentation
- ServiceNow Access Control (ACL) documentation